

The Blocked or Waiting state triggers (I/O requests, event waits) ,Process synchronization foundations (signals, wait queues, and communication) , Normal vs. abnormal termination system calls (e.g., exit(), abort()) ,Resource reclamation and the concepts of Zombie and Orphan processes

Code:

```
#include <stdio.h>

int main() {

    int n = 4;

    int process[] = {1, 2, 3, 4};
    int burst[] = {5, 3, 8, 2};

    int waiting[4];
    int turnaround[4];

    int total_waiting = 0;
    int total_turnaround = 0;
    int total_burst = 0;

    printf("FCFS Scheduling\n\n");

    /* FCFS waiting time */
    waiting[0] = 0;

    for (int i = 1; i < n; i++) {
        waiting[i] = waiting[i - 1] + burst[i - 1];
    }

    /* Turnaround time */
```

```

/* Turnaround time */
for (int i = 0; i < n; i++) {
    turnaround[i] = waiting[i] + burst[i];

    total_waiting += waiting[i];
    total_turnaround += turnaround[i];
    total_burst += burst[i];
}

printf("Process\tBurst\tWaiting\tTurnaround\n");

for (int i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\n",
        process[i],
        burst[i],
        waiting[i],
        turnaround[i]);
}

printf("\nAverage Waiting Time = %.2f",
    (float)total_waiting / n);

printf("\nAverage Turnaround Time = %.2f",
    (float)total_turnaround / n);

```

```

printf("\nCPU Utilization = 100%%");

printf("\nThroughput = %.2f processes/unit time\n",
       (float)n / total_burst);

/* SJF */

int sjf_process[] = {4, 2, 1, 3};
int sjf_burst[] = {2, 3, 5, 8};

int sjf_waiting[4];
int sjf_turnaround[4];

int sjf_total_waiting = 0;
int sjf_total_turnaround = 0;

sjf_waiting[0] = 0;

for (int i = 1; i < n; i++) {
    sjf_waiting[i] =
        sjf_waiting[i - 1] + sjf_burst[i - 1];
}

```

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    for (int i = 0; i < n; i++) {
        sjf_turnaround[i] =
            sjf_waiting[i] + sjf_burst[i];

        sjf_total_waiting += sjf_waiting[i];
        sjf_total_turnaround += sjf_turnaround[i];
    }

    printf("\n\nSJF Scheduling\n\n");

    printf("Process\tBurst\tWaiting\tTurnaround\n");

    for (int i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            sjf_process[i],
            sjf_burst[i],
            sjf_waiting[i],
            sjf_turnaround[i]);
    }

    printf("\nAverage Waiting Time = %.2f",
        (float)sjf_total_waiting / n);

    printf("\nAverage Turnaround Time = %.2f\n",
        (float)sjf_total_turnaround / n);
}
return 0;
}

```

Output:

```
geya@LenovoL480:~/OS/Practical/Week 6$ ./task2
FCFS Scheduling
```

Process	Burst	Waiting	Turnaround
P1	5	0	5
P2	3	5	8
P3	8	8	16
P4	2	16	18

Average Waiting Time = 7.25
Average Turnaround Time = 11.75
CPU Utilization = 100%
Throughput = 0.22 processes/unit time

SJF Scheduling

Process	Burst	Waiting	Turnaround
P4	2	0	2
P2	3	2	5
P1	5	5	10
P3	8	10	18

Average Waiting Time = 4.25
Average Turnaround Time = 8.75

```
geya@LenovoL480:~/OS/Practical/Week 6$ |
```